Analytical Chemistry Cumulative Exam
September 30, 2011

(At the end of this cumulative exam are equations and constants that might be helpful in answering these questions)

Questions on this analytical chemistry cumulative exam are based on the following research article and the fundamental science and understanding behind the studies described in it:

Kamlesh D. Patel, Anton D. Jerkovich, Jason C. Link, and James W. Jorgenson, "In-Depth Characterization of Slurry Packed Capillary Columns with 1.0-μm Nonporous Particles Using Reversed-Phase Isocratic Ultrahigh-Pressure Liquid Chromatography", *Anal.Chem.* **2004**, *76*, 5777-5786.

1. (7 points) What do the acronyms UHPLC and HPLC stand for?

2. (18 points) List and compare three ways in which UHPLC differs from HPLC, in terms of experimental setup, conditions, or chromatographic results. Wherever possible, use numbers as part of your comparisons (i.e. be quantitative if at all possible).

3. (25 points) What is the Van Deemter equation? Describe the components of the equation and what they mean. What parameters are related to each other in that equation? Draw a Van Deemter plot for larger and for smaller particle size packing material to show how the effect of particle diameter plays a role in column efficiency. Why does particle size affect the Van Deemter plot in this way?

4. (10 points) In the article, several factors were considered in order to understand their affect on the efficiency of UHPLC. Below is a list of those factors. Pick ***one*** and give the argument (including any evidence) about whether it does or doesn't play a significant role in determining the H (height of a plate) in UHPLC.

a. frictional (or flow-induced) heating
b. curvature of the wall of the column
c. "diffusional relaxation" with respect to the inner diameter of columns

5. (20 points) Answer ***one*** of the following questions about some details discussed in the article.

(a) What are "extracolumn effects" and what is it that they influence? How were they experimentally determined by the investigators?

(b) Compare diathermic and adiabatic conditions and explain why they were considered in experiments described in the article. How do the different conditions affect H and k'?

(c) Why was it important to the investigators to do the following, (i) improve the packing procedure and process, as well as (ii) design and construct an improved column fitting? Name one improvement made for each of these issues (one for (i) and one for (ii)).

6. (20 points) Answer ***one*** of the following questions about some useful methods that were described in article.

(a) How were statistical moments used by the investigators? What is a statistical moment? To what commonly known parameters are the zeroth, first, and second statistical moments equal? What advantage does iterative statistical moments have over Gaussian fit and full-width at half-maximum methods?

(b) Cross sections of the column used for UHPLC in the article were evaluated by allowing a beam of electrons to hit the cross section of the column in a vacuum environment. What purpose does the electron beam serve and what was measured? Name this instrumental method.

Equations that might or might not be useful for this Analytical Chemistry Cume

$$H = A + B/u + (C_{MP} + C_{SP})u$$

$$h = H/d_p$$

$$\nu = ud_p/D_m$$

$$H_{calc} = (\sigma_{Det2}^2 - \sigma_{Det1}^2)L/(t_{r(Det2)} - t_{r(Det1)})^2$$

$$\text{power} = F\Delta P$$

$$t_D = r^2/2\gamma D_m$$

$$\Phi = \frac{\Delta P d_p^2}{uL\eta\epsilon_i}$$

$$k' = K(V_s/V_m)$$